Practice workbook · Explain the effects and dangers of alcohol and other drugs

Alcohol slowing reactions and driving, its absorption and liver damage; dangers of cocaine and other drugs; caffeine and painkiller data on speed of effect.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

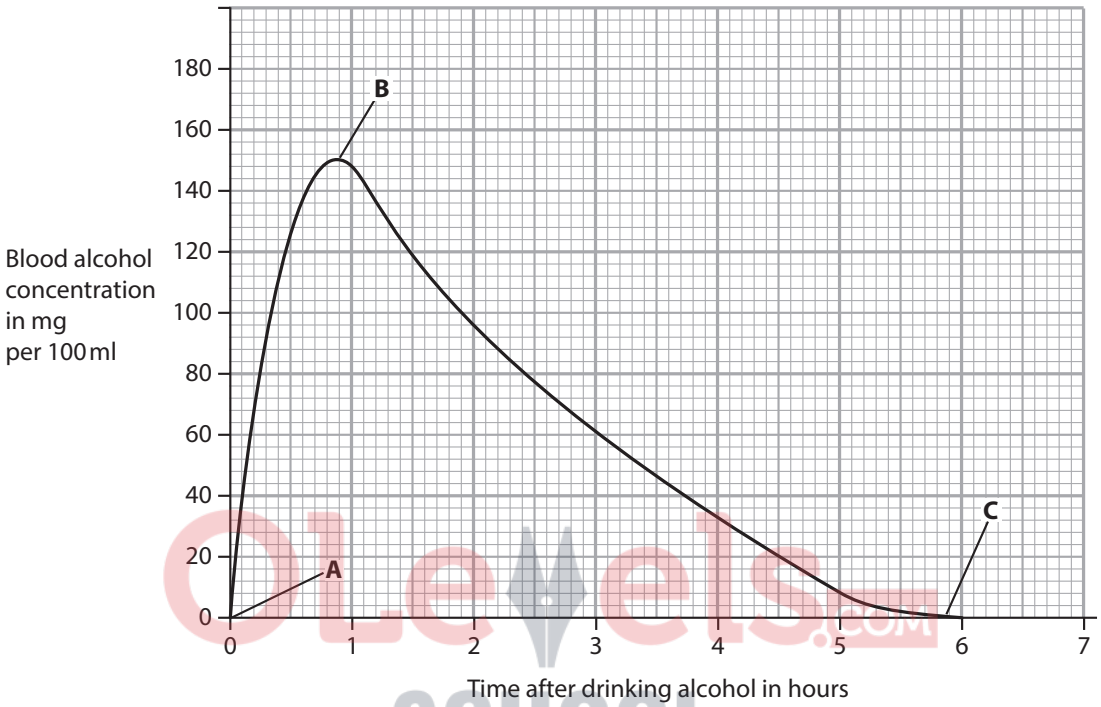
Type ID: HBI-EXPLAIN-DRUG-AND-ALCOHOL-EFFECTS

1. Question 6 · part b.ii · 4 marks

June 2025 Paper 1

Worked-example retry · Attempt part b.ii only.

6 The graph shows the changes in blood alcohol concentration of a person who drinks alcohol.



(a) (i) Explain the shape of the graph between points A and B.

(2)

(ii) Explain the shape of the graph between points B and C.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

1. Question 6 · part b.ii · 4 marks

June 2025 Paper 1

Worked-example retry · Attempt part b.ii only.

(b) In the United Kingdom the legal limit to drive is less than 80 mg per 100 ml of alcohol in the blood.

(i) The person shown in the graph drinks alcohol and the alcohol level in the blood goes above the legal limit to drive.

Determine how many hours after drinking alcohol it would be legal for the person shown in the graph to drive.

(1)

(ii) Explain why it is against the law to drive with high levels of alcohol in the blood.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 6 · part b.ii · 4 marks

June 2025 Paper 1

Worked-example retry · Attempt part b.ii only.

(c) Alcohol reduces the production of ADH.

Explain the effect on the urine produced by a person drinking alcohol.

(5)

(Total for Question 6 = 15 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WR

OLevels.com SCHOOL

1. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 7 · part b.ii · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.ii only.

7 The volume of alcohol in alcoholic drinks can be measured in units.

One unit is equivalent to 10 cm^3 of pure alcohol.

The diagram shows a bottle of wine with 14% alcohol content and a glass containing 250 cm^3 of this wine.



250 cm^3 glass of Ruby Red wine

(a) Calculate the number of units of alcohol in the glass of wine.

(2)

number of units = ...

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

2. Question 7 · part b.ii · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.ii only.

(b) It is illegal for a person to drive a motor vehicle if they have more than the legal limit of alcohol in their bloodstream.

It takes one hour for the body to process one unit of alcohol so that there is no alcohol left in the bloodstream.

(i) A person drinks three units of alcohol. This puts her at the legal limit.

Calculate the time it would take before a person can legally drive after drinking four glasses of Ruby Red wine.

(2)

time = ... hours

(ii) Suggest why it is illegal for people to drive a motor vehicle with more than the legal limit of alcohol in their bloodstream.

(2)

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

O Levels .COM SCHOOL

2. Question 7 · part b.ii · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.ii only.

(c) Drinking a lot of alcohol over a long period of time can cause a condition called cirrhosis.

This condition prevents the cells of the liver from functioning normally.

Explain the effect of cirrhosis on the digestion of food molecules.

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 7 = 11 marks)

2. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 9 · part a · 3 marks

June 2022 Paper 2

Further practice · Attempt part a only.

9 (a) Hepatitis is a disease of the liver.

This disease can be caused by drinking too much alcohol.

Explain the damaging effects of alcohol on the liver.

(3)

(b) Hepatitis B is one type of virus that can cause hepatitis.

Describe the reproduction of a hepatitis B virus.

(6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

3. Question 9 · part a · 3 marks · Working and answer

Working and answer

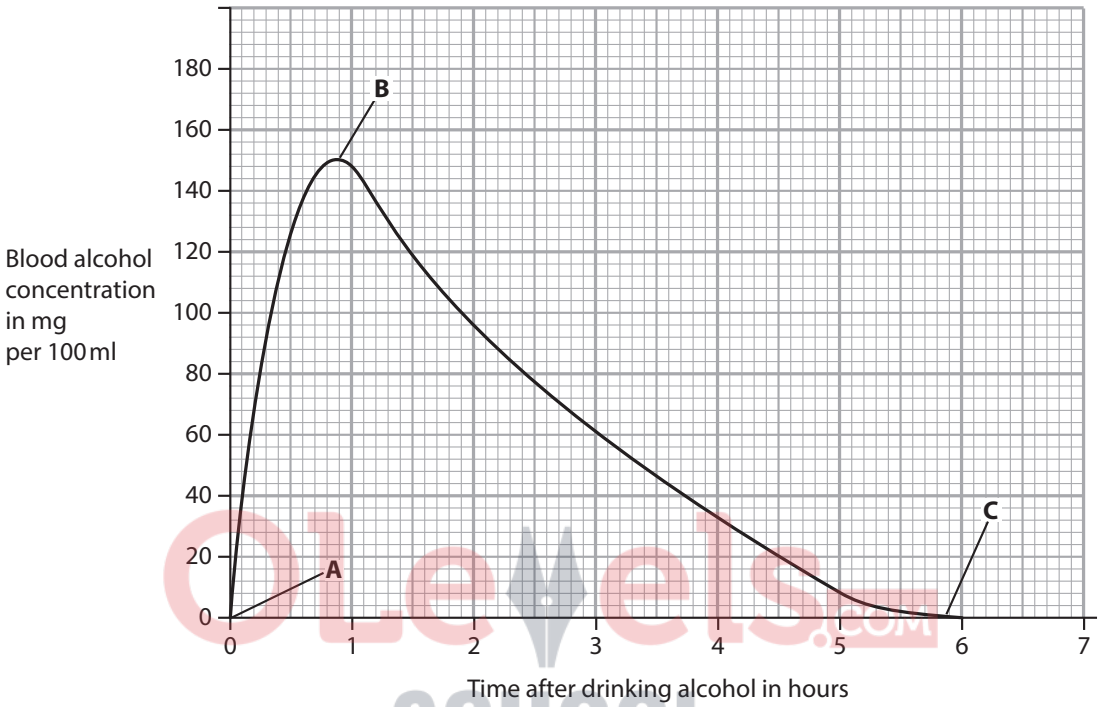
OLevels.com SCHOOL

4. Question 6 · part a.i · 2 marks

June 2025 Paper 1

Further practice · Attempt part a.i only.

6 The graph shows the changes in blood alcohol concentration of a person who drinks alcohol.



(a) (i) Explain the shape of the graph between points A and B.

(2)

(ii) Explain the shape of the graph between points B and C.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 4 · part b · 3 marks

November 2020 Paper 2

Further practice · Attempt part b only.

4 Aspirin is the active ingredient used in some tablets to relieve pain.

Aspirin becomes acidic when it dissolves in the body. The faster aspirin dissolves, the faster it can relieve pain.

A student investigates if it is better to mix sucrose or starch with the active ingredient to make an aspirin tablet.

This is the student's method.

- add distilled water to three beakers, A, B and C
- add 1 aspirin tablet to each beaker
- add 5 g of sucrose to beaker B
- add 5 g of starch to beaker C
- measure the pH of the contents of each beaker at the start of the investigation
- measure the pH of the contents of each beaker every minute for 5 minutes

The table shows the student's results.

Beaker	pH of contents of beaker					
	at start	1 minute	2 minutes	3 minutes	4 minutes	5 minutes
A – aspirin	7.6	7.1	6.6	6.2	5.8	5.4
B – aspirin and sucrose	7.2	6.7	6.3	5.7	5.2	4.7
C – aspirin and starch	7.1	7.2	7.3	7.3	7.4	7.4

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5. Question 4 · part b · 3 marks

November 2020 Paper 2

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) (i) Plot a graph of the student's results.

Join the points with straight lines.

(5)



(ii) State two variables that should be controlled in this investigation.

(2)

1 .....

2 .....

(iii) Name the dependent variable in this investigation.

(1)

.....

5. Question 4 · part b · 3 marks

November 2020 Paper 2

Further practice · Attempt part b only.

(b) Explain how the results of this investigation could help a manufacturer improve the effectiveness of its aspirin tablets.

(3)

DO NOT WRITE IN THIS AREA

(Total for Question 4 = 11 marks)

OLevels.com SCHOOL

Working and answer

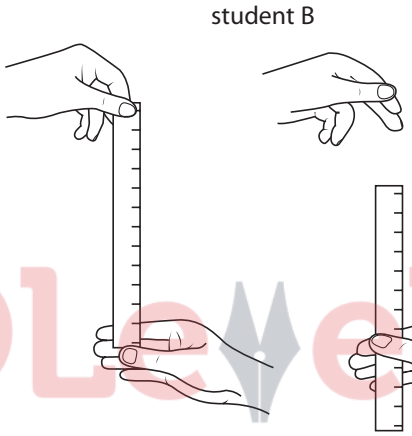
6. Question 5 · part b · 1 mark

November 2021 Paper 2

Further practice · Attempt part b only.

5 Two students use this method to investigate if caffeine affects the speed of a reflex action.

- student A sits in a chair holding out a hand with their thumb and forefinger open
- student B holds a metre rule with the 0 mm mark just above the open thumb and forefinger of student A
- student B releases the ruler
- student A catches the ruler as quickly as possible by closing their thumb and forefinger



OLevels.COM SCHOOL

The method is repeated 10 times.

The student catching the ruler now drinks a cup of coffee and waits for 30 minutes.

The investigation is then repeated.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6. Question 5 · part b · 1 mark

November 2021 Paper 2

Further practice · Attempt part b only.

The table shows the student's results.

Conditions \ Trial	Distance travelled by ruler in mm										
	1	2	3	4	5	6	7	8	9	10	mean
without coffee	210	190	200	205	207	195	198	195	199	201	200
with coffee	165	160	163	167	165	170	175	170	180	175	

(a) Complete the table by calculating the missing mean value.

(2)

(b) Coffee contains caffeine.

State the effect of caffeine on the reflex action.

(1)

(c) Explain why the student waited 30 minutes after drinking the coffee before continuing with the investigation.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 5 · part b · 1 mark · Working and answer

Working and answer

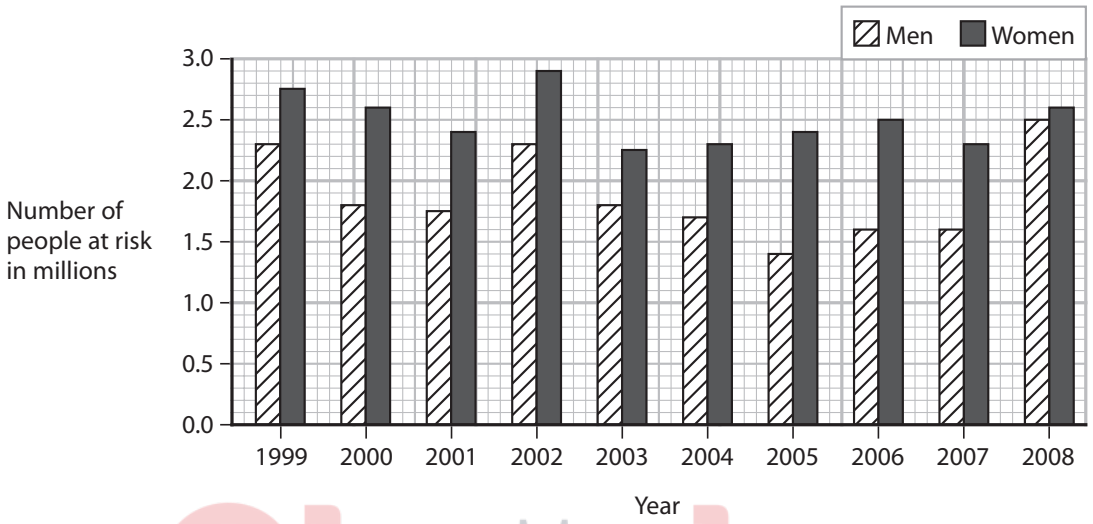
OLevels.com SCHOOL

7. Question 2 · part c · 1 mark

June 2022 Paper 2

Further practice · Attempt part c only.

2 The graph shows the number of working-age people at risk of developing a mental illness.



(a) (i) Determine the year when the greatest number of women were at risk of developing a mental illness.

(1)

.....

(ii) Determine the year when there was the greatest difference between the number of males and the number of females at risk of developing a mental illness.

(1)

.....

(iii) Determine the ratio of females at risk of developing a mental illness in 1999 compared with 2008.

Give your answer in the form n : 1

(2)

ratio = ...

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 0 9 7 4 A 0 4 2 0

7. Question 2 · part c · 1 mark

June 2022 Paper 2

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

(iv) Suggest one possible reason why the number of females at risk of developing a mental illness increased from 2007 to 2008.

(1)

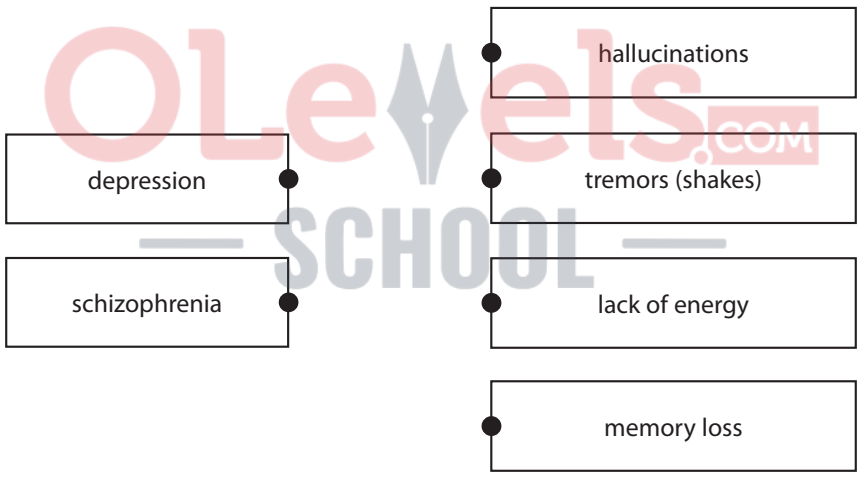
.....

.....

(b) The boxes list two mental illnesses and some symptoms.

Draw one straight line from each mental illness to a symptom of the illness.

(2)



(c) Which combination of drugs is most likely to affect mental health?

(1)

- ☒ **A** alcohol and heroin
- ☒ **B** antibiotics and alcohol
- ☒ **C** cannabis and paracetamol
- ☒ **D** paracetamol and antibiotics

(d) People who use the drug cocaine often have a raised body temperature.

Name two body structures that help to control body temperature.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

7. Question 2 · part c · 1 mark · Working and answer

Working and answer

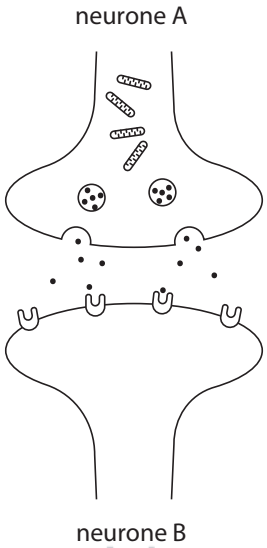
OLevels.com SCHOOL

8. Question 3 · part b · 2 marks

November 2024 Paper 2

Further practice · Attempt part b only.

3 (a) The diagram shows the gap between two neurones, A and B.



(i) Name this gap. **(1)**

(ii) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 5 9 5 9 A 0 8 2 4

8. Question 3 · part b · 2 marks

November 2024 Paper 2

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(iii) Describe how a nerve impulse travels from one side of the gap to the other.

(6)

OLevels.com SCHOOL

(iv) Explain why an impulse can only travel in one direction across the gap.

(3)

P 7 5 9 5 9 A 0 9 2 4

8. Question 3 · part b · 2 marks

November 2024 Paper 2

Further practice · Attempt part b only.

(b) Some people take the illegal drug known as cocaine.

State two dangers of taking cocaine.

(2)

1 .....

.....

2 .....

.....

(Total for Question 3 = 13 marks)

DO NOT WRITE IN THIS AREA

Working and answer

OLevels.com SCHOOL